

Backend feature scope for Tesseract (clean + structured):

CORE IDEA - As we discussed earlier.

Tesseract = IITM society platform (not just games)

Members join → participate in events + games → everything tracked

1. USER ROLES SYSTEM

Roles:

- "guest" (logged in, not joined)
- "member" (joined Tesseract)
- "core" (team member)
- "admin"

2. AUTH SYSTEM (STRICT)

Allowed domains (IIT BS Only)

- @ds.study.iitm.ac.in
- @es.study.iitm.ac.in

Features:

- Email + OTP login
- Domain validation (strict)
- No external users

3. MEMBERSHIP SYSTEM (CORE)

Flow:

Signup → Verify → "guest"
→ Join Tesseract → "member"

Features:

- Membership status
- Access control based on role

4. CORE TEAM SYSTEM

- Admin assigns "core" role
- Separate privileges:
 - Manage events

- Manage games
- View analytics

5. USER DASHBOARD DATA APIs

Return:

- Joined events
- Played games
- Scores
- Activity history

6. EVENT SYSTEM (PRIORITY)

Features:

- Create event (core/admin)
- Join event (members)
- Track participants
- Event history

APIs:

- Create event
- Get events
- Join event
- Get participants

7. GAMES SYSTEM

Features:

- Game registry (dynamic)
- Game config (id, rules, type)
- Score submission
- Track plays

APIs:

- Get games
- Submit score
- Get game data

8. LEADERBOARD SYSTEM

Features:

- Game-wise leaderboard
- Global leaderboard
- Ranking logic

9. ACTIVITY TRACKING (IMPORTANT)

Track:

- Membership joins
- Event participation
- Game plays
- Scores

10. ADMIN SYSTEM

Features:

- Manage users
- Assign roles (core/admin)
- Create/edit/delete events
- Manage games
- View activity logs

11. API STRUCTURE

- /auth
- /members
- /users
- /events
- /games
- /leaderboard
- /admin
- /activity

12. DATABASE COLLECTIONS

- Users
- Members
- Events
- Participants
- Games
- Scores
- ActivityLogs

PHASE 2 (KEEP IN MIND) if time allows you can do lakshay as you already made it up.

- Email system (event mails, welcome)
- Certificates system

- Gamification (points, badges)
- Real-time (live leaderboard)
- Analytics dashboard

IMPORTANT NOTES

- Platform is CLOSED (IITM only)
- Membership is mandatory entry point
- Core team access is admin-controlled
- Backend should be modular + scalable

Frontend team will directly plug into your APIs.

Rest from your end of you want to enhance it go ahead.